

FlexWATCH®
Programmer's Guide
HTTP-Interface Specification for FlexWATCH®

Version 2.0

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Overview

This document specifies HTTP interfaces for FlexWATCH™ server to help developers who develop Web-based application program based on FlexWATCH™ Network cameras and Video servers.

The HTTP CGI API documents VER 2.0 is for firmware version 3.0 or above of any FlexWATCH™ Camera or video server. Thus, you must match firmware version of FlexWATCH™ Camera or video server to use this version of documentation.

HTTP CGI API ver. 1.1 was upgraded to ver. 2.0, since there are lots of functional changes in the the firmware of FlexWATCH™ Network camera and Video servers. Thus, if you already developed Application program based on HTTP CGI API ver. 1.0 or 1.1 and want to use firmware ver. 3.0 of FlexWATCH™ Network camera and Video servers, you are strongly required to update your code following new HTTP CGI API.

1. fwsysget.cgi

→ Request server's information such as firmware version and model.

Query Syntax :

```
http://<server_address>/cgi-bin/fwsysget.cgi?  
FwModId=<fwmod_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID.
- <cgi_version> : CGI interface version(=0x0001)

Example :

```
http://10.10.10.10/cgi-bin/fwsysget.cgi?FwModId=0&FwCgiVer=0x0001
```

Response :

```
Model=0x101c  
KernelVersion=0x0300  
OsType=LINUX  
TimeServer=disconnected  
VideoFormat=0x00000000  
SupportCameraModel="NTSC"  
TimeZone="+ 00:00"  
Time="16:03:06"  
Date="1970/01/01"  
SupportCameraCount=4  
SupportDigitalOutputCount=4  
SupportDigitalInputCount=4  
SerialNo="00:30:6f:88:00:ae"  
FwCgiVer=0x0001
```

Description :

- Model : Information of FlexWATCH™ Model.
1 Channel Camera model = 0x1011

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- 1 Channel Video Server model = 0x1019
- 2 Channel Video Server model = 0x101a
- 4 Channel Video Server model = 0x101c
- 4 Channel Video Recoder model = 0x109c
- KernelVersion : FlexWATCH™ firmware version
- RunTimeServer : Connection status of Time Server and FlexWATCH™ Server
 - true : FlexWATCH™ sever connect time server
 - false : FlexWATCH™server disconnect time server
- TimeServer: Connection status of Time server
 - Connection or disconnection
- TimeZone : Information of Time Zone in which FlexWATCH™ server is located.
 - This value is used to obtain remote server time from local client program.
 - Time calculation method is described at "User's manual".
- Time : FlexWATCH™ server's remote time.
- SupportCameraCount : Number of Input Port.
- SupportDigitalOutputCount : Number of Digital Output(DO).
- SupportDigitalInputCount : Number of Digital input(DI)
- SupportCameraModel : Input Video format of FlexWATCH™ server.
 - NTSC : NTSC TV System (525 - 60 Hz)
 - PAL : PAL TV System (625 - 50 Hz)
- SerialNo: Serial number of FlexWATCH™ server.
 - Serial number is Unique. If it is not, unexpected event will happen.
- FwCgiVer : CGI interface version(=0x0001)

2. fwstream.cgi

➔ Request to get JES Image stream from the server.

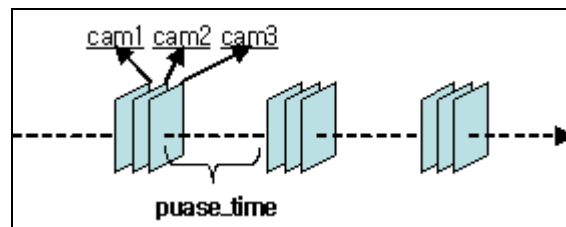
Please refer to 23.1 for getting detailed information of JES(JPEG Extended Stream)

Query Syntax :

```
http://<server_address>/cgi-bin/fwstream.cgi?  
FwModId=<fwmod_id>&  
PortId=<port_id>&  
PauseTime=<pause_time>&  
AppKey=<key>&  
HeaderMode=<header mode>&  
BoundaryType=<boundary type>&  
MuteFlag=<mute flag>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID.
- <port_id> : Port ID <0~3>
- <pause_time> : It is delay time between transmitted stream images.
- Unit is 10ms, the lowest is 0 and Pause time range is <0 ~ n>.



[Pic. 1] Concept of 'Pause_time'

Example : If pause time is 10, $10 \times 10\text{ms} = 100\text{ms}$. This means image can be transmitted every 0.1 seconds. If you want to send image per second, pause time should be 100.

- <app_key> : application key (0x???????) (32 bit hex)
Application key is User Definable in 32 bit hex code.
This value is to prevent unwanted video access such as video hacking.
- Client can use optional value. But this value must be same that is used for

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"fwstctr.cgi".

- <header mode> : Option
0 (Default) : Normal Mode [JES Header + JPEG Image]
1: Header Only Mode [JES Header]
- <boundary type>: Option
0 (Default) : FlexWATCH Standard Boundary Mode
1: --myboundary + Content-Type + Content-Length
- <mute flag>: Option Mute Flag
0 (Default) : Normal Mode
1: Mute mode
- <cgi_version> : CGI interface version(=0x0001)

Example :

When you want to ask JES streams of Video 1, 2 and 3, you can use query syntax as below. (Delay time is 100ms)

http://<server_address>/cgi-bin/fwstream.cgi?

AppKey=0x0450f000&

PortId=0,1,2&

PauseTime=10&

FwCgiVer=0x0001

Response :

HTTP/1.1 200 OK\r\n

Date: Mon, 11 Dec 2000 17:15:55 GMT\r\n

Server: GoAhead-Webs\r\n

Content-Type: multipart/x-mixed-replace

boundary=myboundary\r\n\r\n

--myboundary\r\n

Content-type: image/jpeg\r\n

DaemonId:0x00000013\r\n

JES image data

--myboundary\r\n

Content-type: image/jpeg\r\n

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DaemonId: 0x00000013\r\n\r\n

JES image data

Sample Source :

For analysis and use of Option Field and FES included in JES, please refer to /SDK4_0/HTTP CGI 2.0/man/JES Header Format.ppt and /SDK4_0 /CallFwCgi sources.

3. fwstctr.cgi

→ It is to change Frame rate which is inverse proportion.

Also you can ask to add/delete video input on current video stream.

Query Syntax :

http://<server_address>/cgi-bin/fwstctr.cgi?

FwModId=<fwmod_id>&

PortId=<port_id>&

DaemonId=<daemon_id>&

AppKey=<app_key>&

Action=<action>&

PauseTime=<pause_time>&

FwCgiVer=<cgi_version>

※ PortId, PauseTime optional flag

Parameters :

- <fwmod_id>: Registered server's ID.
- < port _id> : Port ID <0~3>
- <daemon_id> : Daemon ID <0x????????> (32bit hex)
Daemon ID value is included in JES stream. Please refer to Appendix for more info about Demon ID. **Please refer to Appendix 1.**
You can get this value from JES stream
- <app_key> : application key <0x????????*> (32 bit hex)
This value must be same with you set before. (Refer to '**2.fwstream.cgi**').
- <action> : Video Add/Remove, and Set Pause Time <Add/Remove/Set>
- <pause_time> : It is delay time between transmitted stream images.
- Unit is 10ms, lowest is 0 and pause time range is <0 ~ n>. (Refer to '**2.fwstream.cgi**' for more about Pause time).
- <cgi_version> : CGI interface version(=0x0001)
-

Example 1 : Add video Camera 4,5 to current stream.

http://10.10.10.10/cgi-bin/fwstctr.cgi?

FwModId=0&

PortId=2,3&

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DaemonId=0x00000013&

AppKey=0x0450f000&

Action=Add&

FwCgiVer=0x0001

Example 2 : Delete video input 4,5 from current stream.

http://10.10.10.10/cgi-bin/fwstctr.cgi?

FwModId=0&

PortId=2,3&

DaemonId=0x00000013&

AppKey=0x0450f000&

Action=Remove&

FwCgiVer=0x0001

Example 3 : Change transmission Pause Time to 100m sec.

http://10.10.10.10/cgi-bin/fwstctr.cgi?

FwModId=0&

DaemonId=0x00000013&

AppKey=0x0450f000&

Action=Set&

PauseTime=10&

FwCgiVer=0x0001

Response :

If it works fine, you cannot see the result.

4. fwcamimg.cgi

→ It can get a JPEG image. That is snapshot, not stream of JPEG Image

Query Syntax :

```
http://<server_address>/cgi-bin/fwcamimg.cgi?  
FwModId=<fwmod_id>&  
PortId=<port_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID
- <port_id> : Port ID <0~3>
- <cgi_version> : CGI interface version(=0x0001)

Example :

```
http://10.10.10.10/cgi-bin/fwcamimg.cgi?  
FwModId=0&  
PortId=0&  
FwCgiVer=0x0001
```

Response :

It transmits an image.

5. fwptzctr.cgi

→ Request to send PTZ control command to target PTZ device

Query Syntax :

Start & Stop Separated Mode request(New Mode)

http://<server_address>/cgi-bin/fwptzctr.cgi?

FwModId=<fwmod_id>&

PortId=<port_id>&

PtzCode=<ptz_coode>&

PtzParm=<ptz_parm>&

RcvData=< receive_count >&

PresetName=<preset_name>&

TxData=<tx_data_array>&

FwCgiVer=<cgi_version>

parameters :

- <fwmod_id>: Registered server's ID
- <port_id> : PTZ equipment ID <0~3>
- <ptz_coode>: Basic PTZ control code
 - Power PTZ control code
 - Extended PTZ control code
 - Special Function control code
- <ptz_parm>: PtzParm(0x ss rr qq pp)
- < receive_count >: YES/NO or 0~64
- < ptz_coode > : Basic PTZ control code
- <preset_name> : Preset Name string [when the <ptz_coode> is PRESET_SET only]
- <tx_data_array > : Transparent TX and RX data array [when the <ptz_coode> is 0x000100FF only : PtzParm is length of the array]
 - EX: Decimal Case : TxData=10,23,123,25,255&PtzParm=5
 - EX: Hexadecimal Case : TxData=0x0A,0x17,7B,0x19,0xff&PtzParm=5

Action	Step mode	Start or Do	Stop	PtzParm= 0xss rr qq pp
DOWN_LEFT	0x00000300	0x00000100	0x00000200	pp = Speed
DOWN	0x00000301	0x00000101	0x00000201	pp = Speed

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DOWN_RIGHT	0x00000302	0x00000102	0x00000202	pp = Speed
LEFT	0x00000303	0x00000103	0x00000203	pp = Speed
HOME	0x00000304	0x00000104		
RIGHT	0x00000305	0x00000105	0x00000205	pp = Speed
UP_LEFT	0x00000306	0x00000106	0x00000206	pp = Speed
UP	0x00000307	0x00000107	0x00000207	pp = Speed
UP_RIGHT	0x00000308	0x00000108	0x00000208	pp = Speed
AUTOPAN_START	0x00000309	0x00000109		
AUTOPAN_STOP	0x0000030A	0x0000010A		
ZOOM_IN	0x0000030B	0x0000010B	0x0000020B	pp = Speed
ZOOM_OUT	0x0000030C	0x0000010C	0x0000020C	pp = Speed
FOCUS_NEAR	0x0000030D	0x0000010D	0x0000020D	pp = Speed
FOCUS_FAR	0x0000030E	0x0000010E	0x0000020E	pp = Speed
PRESET_SET		0x00000111	0x00000211	pp = Preset ID
PRESET_RUN		0x00000112		pp = Preset ID
IRIS_OPEN	0x00000325	0x00000125	0x00000225	pp = Speed
IRIS_CLOSE	0x00000326	0x00000126	0x00000226	pp = Speed

●

● < ptz_code > : Power PTZ control code

Action	Start or Do	PtzParm = 0x ss rr qq pp
POWER_ON	0x0000010F	
POWER_OFF	0x00000110	
AUX_0_ON	0x00000113	
AUX_0_OFF	0x00000114	
AUX_1_ON	0x00000115	
AUX_1_OFF	0x00000116	
AUX_2_ON	0x00000117	
AUX_2_OFF	0x00000118	

● < ptz_code > : Extended PTZ control code

Action	Start or Do	PtzParm = 0x ss rr qq pp
SET_SWING_START	0x00000119	
SET_SWING_STOP	0x0000011A	
SET_HOME_POS	0x0000011B	
OSD_MENU_ON_OFF	0x0000011C	

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GROUP_RUN_CMD	0x0000011D	pp = Group ID
GROUP_STOP_CMD	0x0000011E	pp = Group ID
TOUR_RUN	0x0000011F	pp = Tour ID
TOUR_STOP	0x00000120	pp = Tour ID
SCAN_RUN	0x00000121	pp = Scan ID
SCAN_STOP	0x00000122	pp = Scan ID
PATTERN_RUN	0x00000123	pp = Pattern ID
PATTERN_STOP	0x00000124	pp = Pattern ID

●

- < ptz_code > : Special Function control code

Special Function	Function Code	PtzParm = 0x ss rr qq pp (Hexadecimal)
Pointing Pan Tilt With +- Zoom Ctrl	0x000000FF	pp = Percent X = [-100 ~ 0 ~ 100] qq = Percent Y = [-100 ~ 0 ~ 100] rr = Percent Z = [-100 ~ 0 ~ 100]
Transparent TX and RX Function	0x000100FF	Transparent TX and RX Function User can describe the array of decimal and hexadecimal with `' Note: PtzParm is length of the array
Reserved 2	0x000200FF	
Get PTZ Value	0x000300FF	Return "PTZ_PARM=0x????????;" [Bit31~Bit0] Pan = 0~360 [Bit11~Bit0]12bit(0x000~0xfff:0~4095) Tilt = 0~90 [Bit21~Bit12]10bit(0x000~0x3ff:0~1023) Zoom = Min ~ Max [Bit31~Bit22]10bit(0x000~0x3ff:0~1023) Note: Must be <receive_count> is YES
Set PTZ Value	0x000400FF	PtzParm=0x????????;" [Bit31~Bit0] Pan = 0~360 [Bit11~Bit0]12bit(0x000~0xfff:0~4095) Tilt = 0~90 [Bit21~Bit12]10bit(0x000~0x3ff:0~1023)

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		Zoom = Min ~ Max [Bit31~Bit22]10bit(0x000~0x3ff:0~1023) TxData[0] = Pan Speed TxData[1] = Tilt Speed TxData[2] = Zoom Speed Set Speed Range = [0~254] No Change Speed = 255 TxData=255, 255, 255 Note: Must be <receive_count> is NO
Advanced Get PTZ Value	0x000500FF	PtzParm=0x00000000; Bit Enable Value Bit0:Pan Position / Bit1:Tilt Position Bit2:Zoom Position / Bit3:Pan Speed Bit4:Tilt Speed / Bit5: Zoom Speed RxData[0],RxData[1]: Pan = 0~35999 RxData[2],RxData[3]: Tilt = 0~35999 RxData[4],RxData[5]: Zoom = 0~65535 RxData[6] = Pan Speed [0x00~0x3f] RxData[7] = Tilt Speed [0x00~0x3f] RxData[8] = Zoom Speed [0x00~0x03] Unknown Speed = 0xff Note: Must be <receive_count> is YES Note: RxData =0x??, ~ 0x?? : 9 bytes Note: RxData[1]=MSB / RxData[0]=LSB Note: RxData[3]=MSB / RxData[2]=LSB Note: RxData[5]=MSB / RxData[4]=LSB Note: Tilt 0~9000 : 0' ~ 90' Note: Tilt 35999~27000 : -0.001'~-90'
Advanced Set PTZ Value	0x000600FF	PtzParm=0x00000000; Bit Enable Value Bit0:Pan Position / Bit1:Tilt Position Bit2:Zoom Position / Bit3:Pan Speed Bit4:Tilt Speed / Bit5: Zoom Speed TxData[0],TxData[1]: Pan = 0~35999

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		TxData[2],TxData[3]: Tilt = 0~35999 TxData[4],TxData[5]: Zoom = 0~65535 TxData[6] = Pan Speed [0x00~0x3f] TxData[7] = Tilt Speed [0x00~0x3f] TxData[8] = Zoom Speed [0x00~0x03] Note: No Change Speed = 0xff Note: Must be <receive_count> is NO Note: TxData=0x??, ~ 0x?? : 9 bytes Note: TxData[1]=MSB / TxData[0]=LSB Note: TxData[3]=MSB / TxData[2]=LSB Note: TxData[5]=MSB / TxData[4]=LSB Note: Tilt 0~9000 : 0' ~ 90' Note: Tilt 35999~27000 : -0.001'~-90'
--	--	--

-
- <cgi_version> : CGI interface version(=0x0001)
-

example :  PTZ_ID=1, Sensitivity=5, move to the left start

```
http://10.10.10.10/cgi-bin/fwptzctr.cgi?  
FwModId=0&  
PortId=0&  
PtzCode=0x00000100&  
PtzParm=0x00000005&  
RcvData=NO&  
FwCgiVer=0x0001
```

Response :

If it works fine, you cannot see the result.

6. fwdoctr.cgi

➔ Request to change Digital Output (Relay output device) data.

Query Syntax :

```
http://<server_address>/cgi-bin/fwdoctr.cgi?  
FwModId=<fwmod_id>&
```


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```
DoMask=<do_mask>&  
DoData=<do_data>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID.
- <do_mask> : 1 Byte DO mask data value <0x??> (HEX value)
- <do_data> : 1 Byte DO data value <0x??> (HEX value)
- <cgi_version> : CGI interface version(=0x0001)
-
-

Example :

 Change Digital Output(DO) data to '0x01'.

```
http://10.10.10.10/cgi-bin/fwdoctr.cgi?  
FwModId=0&  
DoMask=0x0f&  
DoData=0x01&  
FwCgiVer=0x0001
```

Response :

If it works fine, you cannot see the result.

7. fwvmdget.asp

- ➔ Request to get Video information about the server such as number of camera, DI/DO, video format, server Id and Module ID.

Query Syntax :

http://<server_address>/asp-get/fwvmdget.asp?FwModId=0

Parameters :

- <fwmod_id>: Registered server's ID.
-

Example :

 To get video mode from server.

http://10.10.10.10/asp-get/fwvmdget.asp?FwModId=0

Response :

```
NUM_OF_CAM = 2;
NUM_OF_DI = 2;
NUM_OF_DO = 1;
VIDEO_FORMAT = 0xa0;
FWMODID      = 0;
VIDEO_SET_MODE=1;
VIDEO_FORMAT_CFG=1;
OPTIONFLAG=0x0000;
CAMCOUNT=2;
CAMNAME0="Camera 1";
IMGSIZE0=3;
CODECTYPE0=2;
INSTALLED0=1
STREAMFORCHANNEL0=2
CAMNAME1="Camera 2";
IMGSIZE1=0;
CODECTYPE1=0;
INSTALLED1=1
STREAMFORCHANNEL1=2
```

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```
CAMINSTFLAG=0x03;  
CAM_STATE=0x03;  
IMGSIZE=3;  
CODECTYPE=2;  
INSTALLED=1;  
TIMESTAMP=0;  
FRAMERATE=0;  
FRAMESKIPCOUNT=0;  
WATERMARKENABLE=0;  
INTERLACETYPE=0;  
AUDIOINSTALLED=0;  
MAXIMGWIDTH=1280;  
MAXIMGHEIGHT=720;
```

Description :

- NUM_OF_CAM : Number of available channel(camera)
- NUM_OF_DI : Number of available DI
- NUM_OF_DO : Number of available DO
- FWMODID : Registered Server ID (It supports Proxy). '0' is local server.
- VIDEO_FORMAT_CFG : Video Format.
NTSC = 1, PAL = 2
- OPTIONFLAG : Value for Image Information which will be sent along with video
such as serial input data, server name, Camera name.
- Below table shows meaning of each bit.
- (You can use above 2 bits at the same time.)

Bit	Description
bit 0	(reserved)
bit 1	Image encryption
bit 2	IP address
bit 3	Server name
bit 4	Channel (Camera) name
bit 5	PPP status (Not used)
bit 6	User defined message
bit 7	UART sensor data

- KEYSTREAM : Key value for encryption. Maximum 8 byte.

8. fwcamget.asp

→ Request to get information about FW-Module.

Query Syntax :

```
http://<server_address>/asp-get/fwcamget.asp?  
FwModId=<fwmod_id>&  
PortId=<port_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID.
- <port_id> : Port ID <0~3>
- <cgi_version> : CGI Interface version(=0x0001)
-

Example :

 To get information of camera. fwmod_id=0, port_id=0

```
http://10.10.10.10/asp-get/fwcamget.asp?  
FwModId=0&  
PortId=0&  
FwCgiVer=0x0001
```

Response :

```
FWMODID=0;  
PORTID=0;  
CAMNAME="Camera 1";  
INSTALLED=1;  
COLORMODE=1;  
QUALITY=3;  
IMGSIZE=1;  
HUE=0;  
SATURATION=0;  
CONTRAST=0;  
BRIGHTNESS=0;
```

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```
SHARPNESS=0;  
SWDELAY=0;  
AUTODETECT=1;  
TIMESTAMP=0;  
MDENABLE=0;  
MDSSENS=0;  
MDMASK="0:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0";
```

Description :

Refer to '14. fwcamset.cgi' for the more about parameters about camera setting.

9. fwdioget.asp

➔ Request to get information about alarm input (DI) and Relay output (DO) device.

Query Syntax :

```
http://<server_address>/asp-get/fwdioget.asp?  
FwModId=<fwmod_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id> : Registered server's ID .
- <cgi_version> : CGI interface version(=0x0001)
-

Example :

 To get information of DI/DO. fwmod_id=0.

```
http://10.10.10.10/asp-get/fwdioget.asp?  
FwModId=0&  
FwCgiVer=0x0001
```

Response :

```
FWMODID=0;  
DICOUNT=0x04;  
DOCOUNT=0x04;  
DI_DATA=0x00;  
DO_DATA=0x00;  
  
FWMODID =0;  
DICOUNT=4;  
DOCOUNT=4;  
DINAME0="Di 1";  
DONAME0="Do 1";  
DINAME1="Di 2";  
DONAME1="Do 2";  
DINAME2="Di 3";  
DONAME2="Do 3";
```

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```
DINAME3="Di 4";  
DONAME3="Do 4";
```

Description :

- FWMODID : It must be Parameter's value.
- DICOUNT : Number of DI port in the Server
- DOCOUNT : Number of DO port in the server
- DO_DATA : Information of DO status.
- It is indicated DO's status from each 'bit 0'.
- It is 'open' if value is 0, and it is 'close' if value is 1.
- DI_DATA : Information of DI status.
- It is indicated DI's status from each 'bit 0'.
- It is 'close' if value is 0, and it is 'open' if value is 1.
- DINAME_n : It is configured DI's name.
- DONAME_n : It is configured DO's name.
-
-

10. fwptzget.asp

→ Request to obtain the parameters of PTZ(Pan/Tilt/Zoom) devices which is installed at target FW server

Query Syntax :

```
http://<server_address>/asp-get/fwptzget.asp?  
FwModId=<fwmod_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id> : Registered server's ID.
- <cgi_version> : CGI interface version (=0x0001)
-

Example :

 It gets a information of PTZ device from fwmod_id 0.

```
http://10.10.10.10/asp-get/fwptzget.asp?  
FwModId=0&  
FwCgiVer=0x0001
```

Response :

```
PTZINSTCOUNT=30;  
PTZPORTCOUNT=4;  
PTZDEVICEID=-1;  
UART_LINE_MODE=0;  
PTZBASEADDR0=0;  
PTZINSTFLAG0=0x00000000;  
PTZBASEADDR1=4;  
PTZINSTFLAG1=0x00000000;  
PTZBASEADDR2=8;  
PTZINSTFLAG2=0x00000000;  
PTZBASEADDR3=12;  
PTZINSTFLAG3=0x00000000;
```

Description :

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- PTZINSTCOUNT : Number of PTZ device supported by 'FW-Module(Respective server)'.
- PTZPORTCOUNT : Number of PTZ port for each 'FW-Module (Respective server)'
- PTZDEVICEID : Vender ID of installed PTZ. If ID is '-1', it means 'Not installed'. The range is 0~(PTZINSTCOUNT-1).
- PTZBASEADDR_n : PTZ Base address. Default is "0". System default PTZ ID for each video port will be changed if PTZ Base address is changed.
- PTZINSTFLAG : To indicate whether or not PTZ is installed at each FW-Module.
-

11. fwusrget.asp

→ fwusrget.asp gets User Property Values.

Query Syntax :

http://<server_address>/asp-get/fwusrget.asp

Parameters :

None

Example :

 It gets property of user.

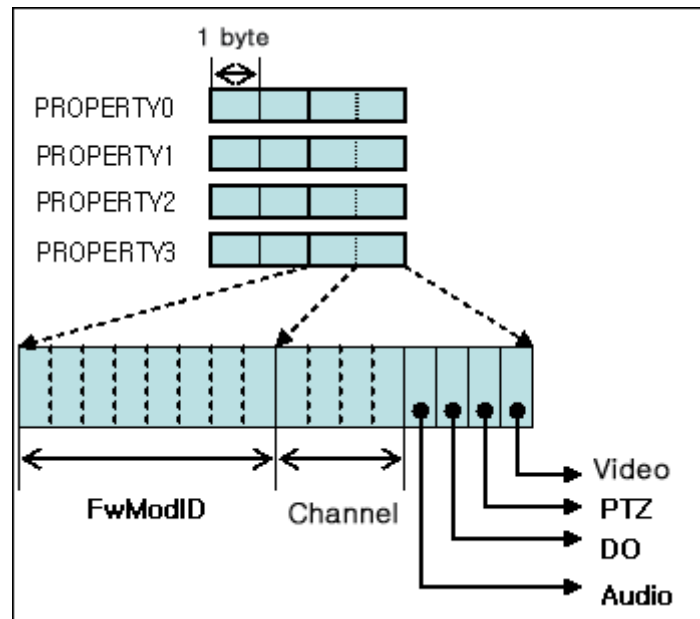
http://10.10.10.10/asp-get/fwusrget.asp

Response :

```
<!-- NssGetUserProp("AuthMode") success -->
USERID="root";
PASSWD="";
FULLNAME="Administrator";
PROPERTY0=0xffffffff;
PROPERTY1=0xffffffff;
PROPERTY2=0xffffffff;
PROPERTY3=0xffffffff;
PROPERTY4=0xffffffff;
PROPERTY5=0xffffffff;
PROPERTY6=0xffffffff;
PROPERTY7=0xffffffff;
```

Description :

- USERID : Authenticated ID.
- PASSWD :
- FULLNAME : Full name of authenticated ID.
- PROPERTY_n : It gets property of user. It uses PROPERTY0 ~ PROPERTY3 which is 4bytes.(PROPERTY4 ~ PROPERTY7 are not used.).
- It has 8 properties like below [Pic. 4].



[Pic. 2] Structure of Property

- '0xFF' means that control authority is given.
- '0x00' means that control authority is not given.
-

12. fwuartget.asp

➔ It gets UART device mode, which are PTZ, Audio, Modem, Console, UART Sensor or UART Output Devices

Query Syntax :

http://<server_address>/asp-get/fwuartget.asp

Parameters :

None

Example :

 To get uart information from the server.

http://10.10.10.10/asp-get/fwuartget.asp

Response :

```
TTYS0_MODE="c";
TTYS1_MODE="p";
TTYS2_MODE="c";
TTYS3_MODE="c";
IDE_MODE=0x00000000;
SYF_STR0="FW-3440-D";
SYF_STR1="27-B1444S2410FF2200000";
SYF_STR2="a_ker_xxa_rfs_x4a_sys_x4a_web_xx";
SYF_STR3="";
SYF_STR4="";
SYF_STR5="";
SYF_STR6="";
SYF_STR7="39E1A85496D52BBC315E420D74B5766D";
```

Description :

- TTYS_n_MODE
 - TTYS0_MODE : COM port
 - TTYS1_MODE : AUX port
 - TTYS2_MODE, TTYS3_MODE : Not used

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Value	Description	Value	Description
'p'	PTZ Device	's'	UART Sensor
'a'	Audio Device	'm'	Modem
'u'	UART Out	'c'	Console

13. fwvmdset.cgi

- ➔ To set up video related configuration in the server such as Video format (NTSC/PAL), Option flag (to use user data field in the image), Image encryption, Encryption key.
FlexWATCH only permit "root" user.

Query Syntax :

```
http://<server_address>/cgi-bin/admin/fwvmdset.cgi?  
FwModId=<fwmod_id>&  
Installed=<installed>&  
ImgSize=<img_size>&  
CodecType=<codec_type>&  
FrameRate=<frame_rate>&  
TimeStamp=<timestamp>&  
VideoFormat=<video_format>&  
OptionFlag=<option_flag>&  
KeyStream=<keystream>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id> : Registered server's ID.
- <installed> : Whether installed or not.
 - 1 : Installed
 - 0 : Not installed
- <img_size> : Image size
(Refer to 23.3. Image Size Code Table)
- <codec_type> : Video encoding and decoding type
 - 0 : M-JPEG
 - 1 : MPEG-4
 - 2 : H.264
- <frame_rate> : Value range [0~6] Encoding frame rate

Value	NTSC (fps)	PAL (fps)
0	30	25
1(*)		

HTTP CGI Interface V2.0

2(*)		
3	15	12.5
4	7.5	6.25
5	3.75	3.13
6	1	1

* Not support

- <frame_rate> : Value range [10~39] Encoding frame rate

Value	NTSC (fps)	Value	PAL (fps)
10	30	10	25
11	15	11	12.5
12	10	12	8.33
13	7.5	13	6.25
15	5	14	5
17	3.75	16	3.57
19	3	19	2.5
24	2	24	1.67
39	1	34	1

- <time_stamp> : Display Time-stamp on the image.
0 : No Time-stamp
1 : Enable Time-stamp
- <video_format> : NTSC or PAL mode.
1: NTSC
2: PAL
- <option_flag> : It means whether encryption of image or not, and contents of user data filed of each image. Bit of ption_flag is like below table. Possible of combination is more than 2 bits.

Bit	Description
bit 0	(Reserved)
bit 1	Image Encryption
bit 2	IP address
bit 3	Server name
bit 4	Channel (camera) name
bit 5	PPP status (Not used)
bit 6	User defined message

HTTP CGI Interface V2.0

bit 7	UART sensor data
-------	------------------

For example, if you want to input IP address and server name to image, you should configure '0x1c' as option_flag.

- <keystream> : It is image encryption key. The key is for image encryption. Maximum is 8 bytes.
- <cgi_version> : CGI interface version(=0x0001)
-

Example :

```
http://10.10.10.10/cgi-bin/admin/fwvmdset.cgi?  
FwModId=0&  
Installed=1&  
ImgSize=1&  
CodecType=1&  
FrameRate=3&  
TimeStamp=0&  
VideoFormat=1&  
OptionFlag=0x1c&  
KeyStream=aaa&  
FwCgiVer=0x0001
```

Response :

If it works fine, you cannot see the result.

Description :

Apply to configuration after rebooting

14. fwcamset.cgi

→ It is used for configuration of camera, after authentication for administrator's ID.

Query Syntax :

```
http://<server_address>/cgi-bin/admin/fwcamset.cgi?  
FwModId=<fwmod_id>&  
PortId=<port_id>&  
CamName=<name>&  
Installed=<installed>&  
ColorMode=<color_mode>&  
ImgSize=<img_size>&  
Quality=<quality>&  
GopValue=<gop_value>&  
EncProfile=<profile>&  
CbrValue=<cbr_value>&  
MaxCvbrRatio=<max_cvbr_value>&  
FrameRate=<frame_rate>&  
AudioInstalled=<audio_installed>&  
Hue=<hue>&  
Contrast=<contrast>&  
Saturation=<saturation>&  
Brightness=<brightness>&  
Sharpness=<sharpness>&  
MdEnable=<md_enable>&  
MdSens=<md_sens>&  
MdMask=<md_mask>&  
TimeStamp=<timestamp>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id> : Registered server's ID.
- <port_id> : Port ID <0~3>
- <name> : Camera name. Maximum is 20 bytes.
- <installed> : Whether installed or not.

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1 : Installed

0 : Not Installed

- <color_mode> : Whether color mode or not.

1 : Color mode

0 : Black and White mode

- <img_size> : Image size.

(Refer to 23.3. Image Size Code Table)

- <quality> : Image Quality

Value	Image Quality
0	Low Compression
1	Highest
2	High
3	Normal
4	Low
5	Lowest

- <gop_value> : GOP value of MPEG-4 streaming. Between 1 ~ 255

- <profile> : Encoding Profile of H.264 (Only H.264 supported products)

* Rev-E, Rev-G Only

Value	Encoding Profile
1	Base
2	Main
4	High

- <cbr_value> : Encoding bit-rate

Value	Description
0	VBR (Encoding bit-rate shall be set according to image quality)
1	CBR (32 Kbps)
2	CBR (64 Kbps)
3	CBR (128 Kbps)
4	CBR (256 Kbps)
5	CBR (512 Kbps)
6	CBR (1.0 Mbps)
7	CBR (1.5 Mbps)
8	CBR (2.0 Mbps)
9	CBR (3.0 Mbps)

HTTP CGI Interface V2.0

10	CBR (4.0 Mbps)
11	CBR (5.0 Mbps)
12	CBR (6.0 Mbps)
13	CBR (8.0 Mbps)
14	CBR (10.0 Mbps)
15	CBR (12.0 Mbps)
33	CVBR (32 Kbps)
34	CVBR (64 Kbps)
35	CVBR (128 Kbps)
36	CVBR (256 Kbps)
37	CVBR (512 Kbps)
38	CVBR (1.0 Mbps)
39	CVBR (1.5 Mbps)
40	CVBR (2.0 Mbps)
41	CVBR (3.0 Mbps)
42	CVBR (4.0 Mbps)
43	CVBR (5.0 Mbps)
44	CVBR (6.0 Mbps)

- <max_cvbr_value> : Available Bit Rate of H.264 (Only H.264 supported products)

* Rev-E, Rev-G Only

Value	Description
2	<cbr_value> + 2 Mbps
3	<cbr_value> + 3 Mbps
4	<cbr_value> + 4 Mbps
5	<cbr_value> + 5 Mbps
6	<cbr_value> + 6 Mbps
7	<cbr_value> + 7 Mbps
8	<cbr_value> + 8 Mbps
9	<cbr_value> + 9 Mbps
10	<cbr_value> + 10 Mbps

- <frame_rate> : Value range [0~6] Encoding frame rate

Value	NTSC (fps)	PAL (fps)
0	30	25

HTTP CGI Interface V2.0

1(*)		
2(*)		
3	15	12.5
4	7.5	6.25
5	3.75	3.13
6	1	1

* Not support

- <frame_rate> : Value range [10~39] Encoding frame rate

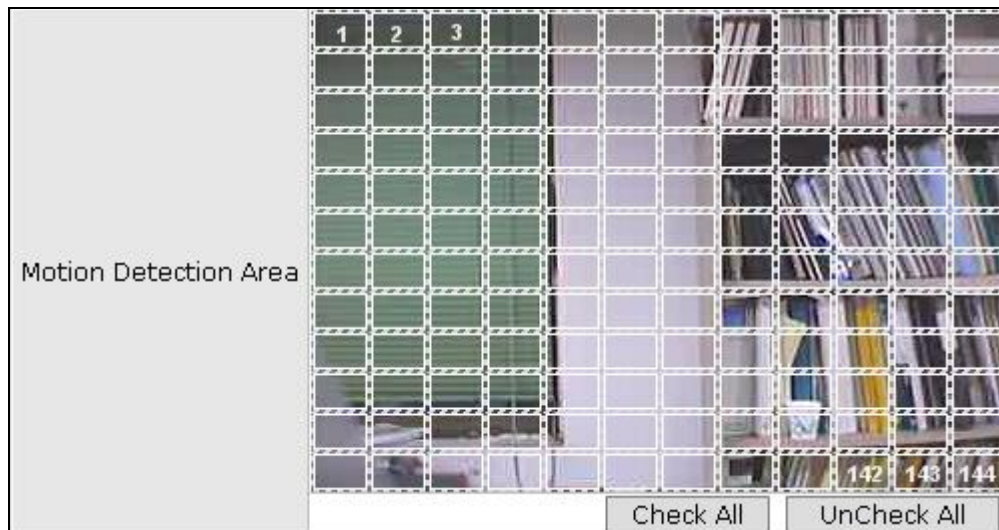
Value	NTSC (fps)	Value	PAL (fps)
10	30	10	25
11	15	11	12.5
12	10	12	8.33
13	7.5	13	6.25
15	5	14	5
17	3.75	16	3.57
19	3	19	2.5
24	2	24	1.67
39	1	34	1

- <audio_installed> : Audio installed or not..
1 : Installed
0 : Not installed
- <hue> : Hue of image. Range is -100~100.
- <contrast> : Contrast of image. Range is -100~100.
- <saturation> : Saturation of image. Range is -100~100.
- <brightness> : Brightness of image. Range is -100~100.
- <sharpness> : Sharpness of image. Range is -100~100.
- <auto_detect> : Whether to use auto detection or not for Video format .
0 : Not use for auto detection.
1 : Auto detection
- <md_enable> : Whether to use motion detection or not supported by FlexWATCH® server.
0 : No motion detection
1 : Motion detection
- <md_sens> : Sensitivity of motion detection. Range is -100~100.

HTTP CGI Interface V2.0

- <md_mask> : Area of motion detection.

Below picture [Pic. 5] shows configuration of motion detection.



[Pic. 3] Area configuration of motion detection

There are 144 image block to screen out unwanted motion area. The count of image block start from top-left to bottom-right in good order.

Value style of <md_mask> is '0:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0'.

Each number of the value is decimal with 8 bits binary(0~255)

Each binary bit mean check or not for motion detection. So Each number represent whether 8 blocks are checked or not, respectively.

It can record whether checked of all box (114 boxes, =18x8) or not from top-left to bottom-right.

For example, if you check just no.1, the value is

'1:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0'.

- <time_stamp> : Whether time records to image or not.
- 0 : No recording.
- 1 : Recording.

- <cgi_version> : CGI interface version(=0x0001)

-

- Example :

<http://10.10.10.10/cgi-bin/admin/fwcamset.cgi?>

FwModId=0&

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PortId=0&
CamName=aaa&
Installed=1&
ColorMode=1&
ImgSize=0x01&
Quality=0x02&
GopValue=4&
EncProfile=4&
CbrValue=0&
MaxCvbrRatio=2&
FrameRate=0&
AudioInstalled=1&
Hue=0&
Contrast=0&
Saturation=0&
Brightness=0&
Sharpness=0&
MdEnable=1&
MdSens=0&
MdMask=128:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0:0&
TimeStamp=0&
FwCgiVer=0x0001

Response :

If it works fine, you cannot see the result.

15. fwdioset.cgi

- → To configure name of DI and DO which installed to server.
-

Query Syntax :

```
http://<server_address>/cgi-bin/admin/fwdioset.cgi?  
FwModId=<fwmod_id>&  
DiName0=<name>&  
DoName0=<name>&  
DiType0=<di_type>&  
DoType0=<do_type>&  
...  
DiNamen=<name>&  
DoNamen=<name>&  
DiTypen=<di_type>&  
DoTypen=<do_type>&  
FwCgiVer=<cgi_version>
```

(*n* is number of DI and DO. It is different from model.)

-

Parameters :

- <fwmod_id> : Registered server's ID.
- <name> : Name of DI and DO. Maximum is 30 bytes, only alphabet.
- <di_type> : Normal Close=0, Normal Open=1.
- <do_type> : Normal Close=1, Normal Open=0.
- <cgi_version> : CGI interface version(=0x0001)

Example :

 Configure a name of the first DI and DO.

```
http://10.10.10.10/cgi-bin/admin/fwdioset.cgi?  
FwModId=0&  
DiName0=door&  
DoName0=window&  
FwCgiVer=0x0001
```

Response :

If it works fine, you cannot see the result.

16. fwptzset.cgi

- → To configure PTZ Model and Install flag in the FW-Module

Query Syntax :

```
http://<server_address>/cgi-bin/admin/fwptzset.cgi?  
PtzDeviceId = <ptzdeviceid>&  
PtzInstFlag0 = <ptzinstflag>&  
PtzInstFlag1 = <ptzinstflag>& (Reserved for multiple video module)  
PtzInstFlag2 = <ptzinstflag>& (Reserved for multiple video module)  
PtzInstFlag3 = <ptzinstflag>& (Reserved for multiple video module)  
PtzBaseAddr0 = <ptzbaseaddr>&  
PtzBaseAddr1 = <ptzbaseaddr>& (Reserved for multiple video module)  
PtzBaseAddr2 = <ptzbaseaddr>& (Reserved for multiple video module)  
PtzBaseAddr3 = <ptzbaseaddr>& (Reserved for multiple video module)  
FwCgiVer=<cgi_version>
```

Parameters :

<ptzdeviceid>: PTZ Device ID
0~(Maximum Install PTZ Count -1)

<ptzinstflag>: Installation Bit Flag
For example,
If PtzInstFlag0=0x00, then it means No PTZ is installed at the Video Module 0.
If PtzInstFlag0=0x05, then it means PTZ are installed at Port ID 2 & 0:
0x05 = b00000101 = channel 3 & 1.

Example :

```
http://10.10.10.10/cgi-bin/admin/fwptzset.cgi?  
PtzDeviceId = 0&  
PtzInstFlag0 = 0x0F&  
PtzInstFlag1 = 0x0F&  
PtzInstFlag2 = 0x00&  
PtzInstFlag3 = 0x00&  
PtzBaseAddr0 = 0&
```


HTTP CGI Interface V2.0

```
PtzBaseAddr1 = 4&
PtzBaseAddr2 = 0&
PtzBaseAddr3 = 0&
FwCgiVer=0x0001
```

Response :

If it works fine, you cannot see the result.

17. fwuartsns.cgi

- → To get the UART Sensor stream data which is received by UART Sensor Module in the FlexWATCH

Query Syntax :

```
http://<server_address>/cgi-bin/fwuartsns.cgi?
FwCgiVer=<cgi_version>
```

Parameters :

- <cgi_version> : CGI interface version(=0x0001)

Example :

```
http://10.10.10.10/cgi-bin/fwuartsns.cgi?FwCgiVer=0x0001
```

Response :

Response data format is binary data stream format. Followed Binary Formatted Header Data after the HTTP Header is Little Endian Number. And the total size is 44 bytes. After that, the UART Data Stream is attached. Among the 44 bytes Header, "UART Data Stream Size" field indicate UART Data Stream size with Auxiliary UART Data Header 4 bytes which is started [0x55, 0xAA].

```
HTTP/1.1 200 OK\r\n
Date: Mon, 11 Dec 2000 17:15:55 GMT\r\n
Server: GoAhead-Webs\r\n
Content-Length: %d\r\n
Content-Type:multipart/formed-data\r\n\r\n\r\n
```

4 th Byte	3 rd Byte	2 nd Byte	1 st Byte	Offset
----------------------	----------------------	----------------------	----------------------	--------

HTTP CGI Interface V2.0

Jiffies Time Stamp(100 Hz)				0
1 st Parsed Sensor Data from UART Data Stream				4
2 nd Parsed Sensor Data from UART Data Stream				8
3 rd Parsed Sensor Data from UART Data Stream				12
4 th Parsed Sensor Data from UART Data Stream				16
Followed Frame Length				20
Media ID	Camera ID	Module ID		24
Received Time				28
Event ID				32
UART Data Stream Size (Include Header Size:4)				36
Reserved	Minor ID	0xAA	0x55	40
.....			UART Data	44

18. fwprearm.cgi

➔ Request to get JES Image stream from the server.

Please refer to 23.1 for getting detail information of JES(JPEG Extended Stream)

Query Syntax :

```

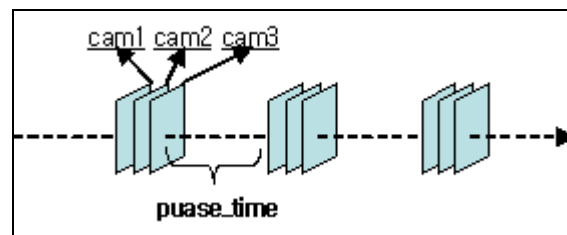
http://<server_address>/cgi-bin/ fwprearm.cgi?
FwModId=<fwmod_id>&
PortId=<port_id>&
PauseTime=<pause_time>&
RmOpt=<remove_option>&
AppKey=<key>&
FwCgiVer=<cgi_version>

```

Parameters :

- <fwmod_id>: Registered server's ID.
- <port_id> : Port ID <0~3>
- <pause_time> : It is delay time between transmitted stream images.
- Unit is 10ms, lowest is 0 and Pause time range is <0 ~ n>.

HTTP CGI Interface V2.0



[Pic. 4] Concept of 'Pause_time'

Example : If pause time is 10, $10 \times 10\text{ms} = 100\text{ms}$. This means image can be transmitted every 0.1 seconds. If you want to send image per second, pause time should be 100.

- `< remove_option >` : Remove Option value for saved pre alarm images `<0/1>`
0: No removing after getting pre alarm images.
1: Removing after getting pre alarm images.
- `<app_key>` : application key (0x????????) (32 bit hex)
Application key is User Definable in 32 bit hex code.
This value is to prevent unwanted video access such as video hacking.
Client can use optional value.
- `<cgi_version>` : CGI interface version(=0x0001)

Example :

When you want to ask JES streams of Video 1,2 and 3, you can use query syntax as below. (Delay time is 100ms)

`http://<server_address>/cgi-bin/ fwprearm.cgi?`

`AppKey=0x0450f000&`

`PortId=0,1,2&`

`PauseTime=10&`

`RmOpt=0&`

`FwCgiVer=0x0001`

Response :

`HTTP/1.1 200 OK\r\n`

`Date: Mon, 11 Dec 2000 17:15:55 GMT\r\n`

`Server: GoAhead-Webs\r\n`

`Content-Type: multipart/x-mixed-replace`

HTTP CGI Interface V2.0

boundary=myboundary\r\n\r\n

--myboundary\r\n

Content-type: image/jpeg\r\n

DaemonId:0x00000013\r\n

JES image data

--myboundary\r\n

Content-type: image/jpeg\r\n

DaemonId: 0x00000013\r\n\r\n

JES image data

19. fwsetmot.cgi

→ Request to force setting Motion data as by this CGI.

Query Syntax :

http://<server_address>/cgi-bin/fwsetmot.cgi?

FwModId=<fwmod_id>&

Motion=< motion_data>&

FwCgiVer=<cgi_version>

Parameters :

- <fwmod_id>: Registered server's ID.
- < motion_data > : 1 Byte Motion data value <0x??> (HEX value)
- <cgi_version> : CGI interface version(=0x0001)
-
-

Example :

☞ To force setting Motion data to '0x01'.

http://10.10.10.10/cgi-bin/ fwsetmot.cgi?

FwModId=0&

Motion=0x01&

HTTP CGI Interface V2.0

FwCgiVer=0x0001

Response :

If it works fine, you cannot see the result.

20. fwaudio.cgi

Request to get JES Audio stream from and to the server(Bidirectional).

Please refer to 20.1 for getting detailed information of JES(JPEG Extended Stream)

Query Syntax:

```
http://<server_address>/cgi-bin/fwaudio.cgi?  
FwModId=<fwmod_id>&  
PortId=<port_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID.
- <port_id> : Port ID <0~3>
- <cgi_version> : CGI interface version(=0x0001)

Example :

 When you want to ask JES Audio streams of port 0, you can use query syntax as below.

```
http://10.10.10.10/cgi-bin/fwaudio.cgi?  
FwModId=0&  
PortId=0&  
FwCgiVer=0x0001
```

Response:

- FW->PC Audio receiving
HTTP/1.1 200 OK\r\n
Server: GoAhead-Webs\r\n
Pragma: no-cache\r\n
Cache-control: no-cache\r\n
Content-Length: 257\r\n

HTTP CGI Interface V2.0

```
Content-Type: text/plain\r\n\r\n
USER_EXT_PROP0=0xffffffff;\n
USER_EXT_PROP1=0xffffffff;\n
USER_EXT_PROP2=0xffffffff;\n
USER_EXT_PROP3=0xffffffff;\n
USER_EXT_PROP4=0xffffffff;\n
USER_EXT_PROP5=0xffffffff;\n
USER_EXT_PROP6=0xffffffff;\n
USER_EXT_PROP7=0xffffffff;\n
AUDIO_INSTALLED=1;\n
AUDIO_CAN_CONNECT=1;\n
AUDIO_2WAY_PERMISSION=1;\n
AUDIO_DEVICE_TYPE=2;\n
--audioboundary\r\n
Content-type: audio\r\n\r\n
```

JES audio data

--audioboundary\r\n
Content-type: audio\r\n\r\n

JES audio data

- PC -> FW Audio sending

Send JES audio data

Send JES audio data

Send JES audio data

Description :

HTTP CGI Interface V2.0

- USER_EXT_PROP *n* : User Permission property. Please refer to 11.fwusrget.asp
- AUDIO_INSTALLED : It is on or off for audio function.
- AUDIO_CAN_CONNECT : It is the status for connection to audio.
- AUDIO_2WAY_PERMISSION : It is the status for bidirectional connection to audio.
- AUDIO_DEVICE_TYPE : Value for Audio codec type.
 - 0 : G723.1
 - Mono,
 - 8K Sampling,
 - Bit Per Sample=8bit
 - Packet Size= 2048 byte
 - 1 : G723
 - Stereo 8K Sampling,
 - Bit Per Sample=16bit
 - Packet Size= 3072 byte
 - 2 : G711
 - u-Low,
 - Mono 8K Sampling
 - Bit Per Sample=8bit
 - PC->FW Packet Size= 1400 bytes
 - FW->PC Packet Size= 4096 bytes
 - 9 : IMA
 - Mono 8K Sampling
 - Bit Per Sample=16bit
 - Packet Size= 2048 bytes

21. fwalarmdaemon.cgi

The internal Event(Status Change) of FlexWATCH will be sent to external client with this CGI connection. Also if external client want to control DO or Recoding, it is used for channel for this control.

'Alarm' is called for the information from the external client to the FlecWATCH and 'Event' is called for the information from the FlecWATCH to the external client.

It is supported after the Firmware Version 4.10

Query Syntax:

HTTP CGI Interface V2.0

http://<server_address>/cgi-bin/fwalarmdaemon.cgi?
FwCgiVer=<cgi_version>

Parameters :

- <cgi_version> : CGI Interface Version(=0x0001)

Example :

Module=0, Port=0 alarm daemon connection

http://10.10.10.10/cgi-bin/fwalarmdaemon.cgi?
FwCgiVer=0x0001

Response:

- FlexWATCH to External Client

event

event

- to External Client to FlexWATCH

alarm

alarm

Sample Source :

- Please refer to /SDK4_0/CallFwCgi source for alarmdaemon connection and data transmission/reception.

Description:

- Protocol Header format (Reference "adp.h" Source code)

Offset Bits	31 ~ 24	23 ~ 16	15 ~ 8	7 ~ 0
0	``	'P'	'D'	'A'

HTTP CGI Interface V2.0

4	Primitive Type (4 Byte Little Endean)	
8	PDU Length (4Byte Little Endean)	
12	Minor Version (2Byte Little Endean)	Major Version (2Byte Little Endean)

- **Primitive Type** description

Primitive Type	description
ADP_ALM_CTRL_REQ	External Client send alarm to Alarm Daemon (FlexWATCH)
ADP_EVT_STAT_IND	Alarm Daemon (FlexWATCH) send event to External Client
ADP_EVT_FES_IND	Alarm Daemon (FlexWATCH) send FES data (POS-Point of Sail, VAMI-Video Analytic Module Interface) Event to External Client

- Alarm PDU description (External Client to FlexWATCH)

Payload data format for alarm control. (Reference "adp.h" Source code)

Offset-Bits	31 ~ 24	23 ~ 16	15 ~ 8	7 ~ 0
16	NVR ID(4 Byte Little Endean) (reserved)			
20	Module ID(4 Byte Little Endean)			
24	Port ID(4 Byte Little Endean)			
28	Group ID(4 Byte Little Endean)			
32	Control Type (4 Byte Little Endean)			
36	Control Parameter[0] (4 Byte Little Endean)			
40	Control Parameter[1] (4 Byte Little Endean)			
44	Control Parameter[2] (4 Byte Little Endean)			
48	Control Parameter[3] (4 Byte Little Endean)			

- Alarm Type (**Control Type**) Field Definition

Alarm Type (Control Type)	Value	Discretion
ALM_CTRL_REC_START	1	Recoding Start Control Command
ALM_CTRL_REC_STOP	2	Recoding Stop Control Command
ALM_CTRL_DO_ON	3	DO On Control Command

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ALM_CTRL_DO_OFF	4	DO Off Control Command
ALM_CTRL_PRESET_RUN	5	Preset Run Control Command (Control Parameter[0]= Preset ID)
ALM_CTRL_POPUP_WND	6	Windows System Program only
ALM_CTRL_TOURING_START	7	Touring Start Command
ALM_CTRL_TOURING_STOP	8	Touring Stop Command

- Event PDU description (FlexWATCH to External Client)

Payload data format for Event status. (Reference "adp.h" Source code)

Offset Bits	31 ~ 24	23 ~ 16	15 ~ 8	7 ~ 0
16	NVR ID(4 Byte Little Endean) (reserved)			
20	Module ID(4 Byte Little Endean)			
24	Port ID(4 Byte Little Endean)			
28	Group ID(4 Byte Little Endean)			
32	State Type (4 Byte Little Endean)			
36	State Parameter[0] (4 Byte Little Endean)			
40	State Parameter[1] (4 Byte Little Endean)			
44	State Parameter[2] (4 Byte Little Endean)			
48	State Parameter[3] (4 Byte Little Endean)			

- Event Type (**State Type**)

Event Type (State Type)	Value	Discretion
EVT_STAT_REC_START	1	Recoding Started Event State Parameter[0] =File ID, State Parameter[1] = File Offset State Parameter[2] = GMT Time
EVT_STAT_REC_STOP	2	Recoding Stop Event
EVT_STAT_MOT_START	3	Motion Start Event
EVT_STAT_MOT_STOP	4	Motion Stop Event

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EVT_STAT_DI_ON	5	DI On Event
EVT_STAT_DI_OFF	6	DI Off Event
EVT_STAT_VIDEO_NORMAL	7	Video Normal Event
EVT_STAT_VIDEO_LOSE	8	Video Lose Event
EVT_STAT_NET_CONN	9	Network Connection Event
EVT_STAT_NET_DISCONN	10	Network Disconnection Event
EVT_STAT_NVR_STARTUP	11	FlexWATCH Started Event
EVT_STAT_NVR_SHUTDOWN	12	FlexWATCH Stop Event
EVT_STAT_TOURING_START	13	PTZ touring Start Event
EVT_STAT_TOURING_STOP	14	PTZ touring Stop Event
EVT_STAT_DO_ON	15	DO Start Event
EVT_STAT_DO_OFF	16	DO Stop Event
EVT_STAT_POPUP_WND	17	Popup event in NVR.
EVT_STAT_ALIVE	18	Alive event with an interval of 5 seconds.
EVT_STAT_FES	19	FES data occurrence event.
EVT_STAT_AUD_DET_START	20	Audio Detection Start Event
EVT_STAT_AUD_DET_STOP	21	Audio Detection Stop Event

- `adp_evt_fes_ind` : FES data Event PDU Header description (FlexWATCH to External Client, Reference source code "adp.h")

Offset Bits	31 ~ 24	23 ~ 16	15 ~ 8	7 ~ 0
16	NVR ID(4 Byte Little Endean) (reserved)			
20	Module ID(4 Byte Little Endean)			
24	Port ID(4 Byte Little Endean)			
28	Group ID(4 Byte Little Endean)			

- `sda_msg_dat_t`: FES data Event PDU Payload description

FES data payload format. (Reference source code "sda_msg.h")

Offset Bits	31 ~ 24	23 ~ 16	15 ~ 8	7 ~ 0
32	Media ID	Camera ID	Server ID (2Byte Little Endean)	

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36	Receive Time(4 Byte Little Endean)			
40	Event ID (4 Byte Little Endean)			
44	Real Log data length + 4byte (4 Byte Little Endean)			
48	Sequence No	Minor ID	Identity (2Byte Little Endean)	
52	Data[3]	Data[2]	Data[1]	Data[0]
56	Data[7]	Data[6]	Data[5]	Data[4]
...	...	...	...	...
284	Data[231]	Data[230]	Data[229]	Data[228]

● Event ID (FES Data)Type Description

Event ID 종류	Value	Description
EVT_CAT_USER	0x870000xx	This type mean the transmitted data is POS data xx mean POS Model ID
EVT_CAT_VAMI	0x880000xx	This type mean the transmitted data is VAMI data xx mean POS vendor ID VAMI Model ID
EVT_CAT_LOG	0x89000000	Not Used
EVT_CAT_AI	0x8Axx0000 0x8A010000 0x8A020000 0x8A030000 0x8A040000 0x8A050000 0x8AFF0000	Represents AI inference result categories as below License Plate Inference Result Object detection inference result Face detection inference result Parking detection inference result Selected Object detection and Tracking result Default OEM AI Model (When using the user's own AI Module)

22. fwalarmstateget.cgi

➔ Request server's information such as DI/DO and motion detection.

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Query Syntax :

```
http://<server_address>/cgi-bin/fwalarmstateget.cgi?  
FwModId=<fwmod_id>&  
FwCgiVer=<cgi_version>
```

Parameters :

- <fwmod_id>: Registered server's ID.
- <cgi_version> : CGI interface version(=0x0001)

Example :

```
http://10.10.10.10/cgi-bin/fwalarmstateget.cgi?FwModId=0&FwCgiVer=0x0001
```

Response :

```
FW_MODULE_ID=0  
NUM_OF_DI=2  
NUM_OF_DO=1  
NUM_OF_CAM=2  
DI_TYPE=0x00  
DO_TYPE=0x00  
DI_STATE=0x03  
DO_STATE=0x00  
MOTION_STATE=0x00  
CAM_STATE=0x03  
FES_STAMP=0x00000000  
FES_DATA0=0x00000000  
FES_DATA1=0x00000000  
FES_DATA2=0x00000000  
FES_DATA3=0x00000000
```

Description :

- FW_MODULE_ID: Registered server's ID
- NUM_OF_DI: Number of DI
- NUM_OF_DO: Number of DO
- NUM_OF_CAM: Number of Camera
- DI_TYPE: Type of DI
0x00: Normal Closed

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- 0x01: Normal Opened
- DO_TYPE: Type of DO
 - 0x00: Normal Opened
 - 0x01: Normal Closed
- DI_STATE: Current state of DI
 - Bit 0 is express of first DI
 - 0x03 is opened first and second DI.
- DO_STATE: Current state of DO
 - Bit 0 is express of first DO
 - 0x00 is opened first and second DO.
- MOTION_STATE: Current motion detection state
 - Bit 0 is express of first camera motion detection
 - 0x00 is all camera don't occurrd motion detection.
- CAM_STATE: Camera install state.
 - 0x03 is express of two camera.
- FES_STAMP: Time stamp of FES data
- FES_DATA: Real FES data.

23. APPENDIX

23.1. JES(JPEG Extended Stream) Header

'JES Header' is the Seyeon Tech's own Header Format, derived and modified from the original JPEG stream. It supports audio-integrated MPEG4 or G.xx. transmission. The following table shows the scheme of 'JES Header'. The byte order is Big Endian in this field

Start Offset	Octet 0	Octet 1	Octet 2	Octet 3
0	Start of Media		0xff	0xfe
4	JES Header Field length		img_pts	
8	img_pts			
12	img_width		img_height	
16	admin seq	fps idx	connection count	
20	group_sequence	watermark_info	header_ver	
24	gmt_time			
28	Module ID(0 Base)		Camera ID(1 Base)	
32	Year(yyyy)		month(1~12)	day(1~31)
36	hour(0~23)	minute(0~59)	second(0~59)	tick(0~99)
40	di_state	camera_state	do_state	motion
44	di_type	Common di/do	Ext state	Ext Data
48	JES Frame Total length (image_size)			
52	mac_address	mac_address	mac_address	mac_address
56	mac_address	mac_address	option_flag	

Start of Media:

In case of JPEG (Image):	[0xFF], [0xD8]
In case of MPEG-4 (Image):	[0x44], [Sequence Number from I-Frame]
In case of H.264 (Image):	[0x45], [Sequence Number from I-Frame]
In case of H.263 (Image):	[0x46], [Sequence Number from I-Frame]
In case of G.723.1 (audio):	[0x90], [Sequence Number]
In case of G.723 (audio):	[0x91], [Sequence Number]

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In case of G.711 (audio): [0x92], [Sequence Number]

In case of IMA (audio): [0x99], [Sequence Number]

0xFFFE: Start of user data in JES

img_pts: Presentation Time Stamp (6bytes)

JES Header Field length: The length of JES Header Field

Module ID: (0~)

Camera ID: (1~6)

gmt_time: GMT Time

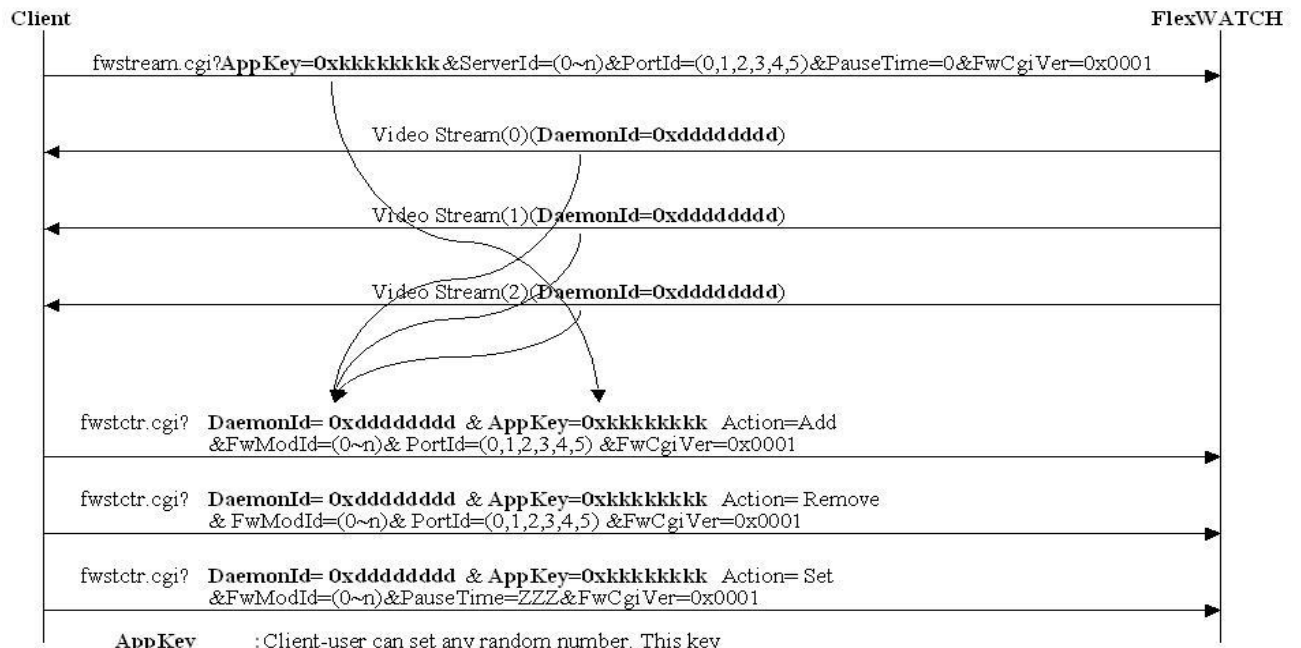
Local Date & Time: Year, Month, Day, Hour, Minute, Sec10mSec

Event Status Field: Di State, CAM State, DO State, Motion State (Each bit is related to port)

JES Frame Total length: Total Length of JES Image (from Start of Media)

23.2. AppKey & DaemonId for fwstream.cgi & fwstctr.cgi

AppKey and DaemonId for fwstream.cgi and fwstctr.cgi V3.0



AppKey : Client-user can set any random number. This key value must be the same as the value used AppKey value at fwstctr.cgi.

DaemonId : Information of DeamonId is included at JPEG stream. This value is allocated by FlexWATCH and it is included boundary information at JPEG stream. All of DeamonId included in the boundary information of JPEG stream are same value for same connection.

When the Pare of **DaemonId** – **AppKey** is correct , the fwstctr.cgi is operated well.

23.3. Image Size Code Table

Each image size code is different size for each product. But you can estimate from the maximum resolution of each product.

So please refer to the following resolution table.

Also you can get the maximum resolution of each product from fwvmdget.asp

Note	VGA	NTSC	PAL	SVGA		XGA	HD 720P
Maximum Resolution	640x480	720x480	720x576	800x600	960x720	1024x768	1280x720
0	160x112	160x112	160x112	160x112	160x112	160x112	160x96
1	320x240	352x240	352x288	320x240	320x240	320x240	320x240
2		704x240	704x288				640x352
3	640x480	704x480	704x576	640x480	640x480	640x480	640x480
4	640x480	720x480	720x576				704x480
5				800x600	960x720	1024x768	
6							1280x720
7							
8							
9							

Note: In the case of Rev-A & Rev-P, Resolution index 1 is 320x240 and 0 is not supported

Note		SXGA		FHD 1080P	QXGA
Maximum Resolution	1280x960	1280x1024	1440x1080	1920x1080	2048x1536
0	160x120	160x112	160x96	160x96	160x112
1	320x240	320x240	320x240	320x240	320x240
2		640x352		640x352	
3	640x480	640x480	640x480	640x480	640x480
4	704x480	704x480	704x480	704x480	704x480
5					
6	1280x960		1440x1080	1280 x 720	
7		1280x1024		1280x1024	1280x1024
8					
9				1920x1080	1920x1080
10					2048x1536

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Note: The real resolution is 1920x1072 or 1920x1080 of the resolution 1920x1080.

Note	4M-1:1	4M-4:3	4M-16:9	5M-4:3	4K
Maximum Resolution	2048x2048	2592x1944	2688x1520	3072x2048	4096x2160
0					160x112
1		320x240	320x240	320x240	256x144
2	320x320				320x240
3	640x640	640x480	640x480	640x480	640x480
4		704x480	704x480	704x480	704x480
5	704x704				
6	1024x1024		1280x720	1280x720	
7		1280x1024	1280x1024	1280x1024	1280x1024
8					
9	1280x1280	1920x1080	1920x108	1920x1080	1920x1080
10		2048x1536		2048x1536	2048x1536
11					
12				1592x1944	
13					
14					
15	2048x2048	2592x1944	2688x1520	3072x2048	2592x1944
16					
17					
18					3456x1944
19					
20					
21					
22					
23					3456x2304
24					
25					
26					3408x2432
27					
28					3840x2160

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29					
30					
31					4096x2160

Note: In the case of Rev-A & Rev-P, Resolution index 1 is 320x240 and 0 is not supported